UCRI 2.0- P2P (Peer-to-Peer) FlowDiagram

Goal

As a **sending system (IPL)**, I want to send messages to a **remote UCRM** via an **Adapter Integration**, so that the target control room system can receive, validate, and process the information according to the UCRI-2.0 P2P(Peer-to-Peer)

Functional Description

Actors

- **Sender System (IPL)**
  Originates business payloads representing events.
- **Adapter Integration**
  Serves as an integration component between IPL and UCRM, responsible for request validation, P2P authentication, message forwarding, and response handling.
- **Target System (UCRM)**

General Flow

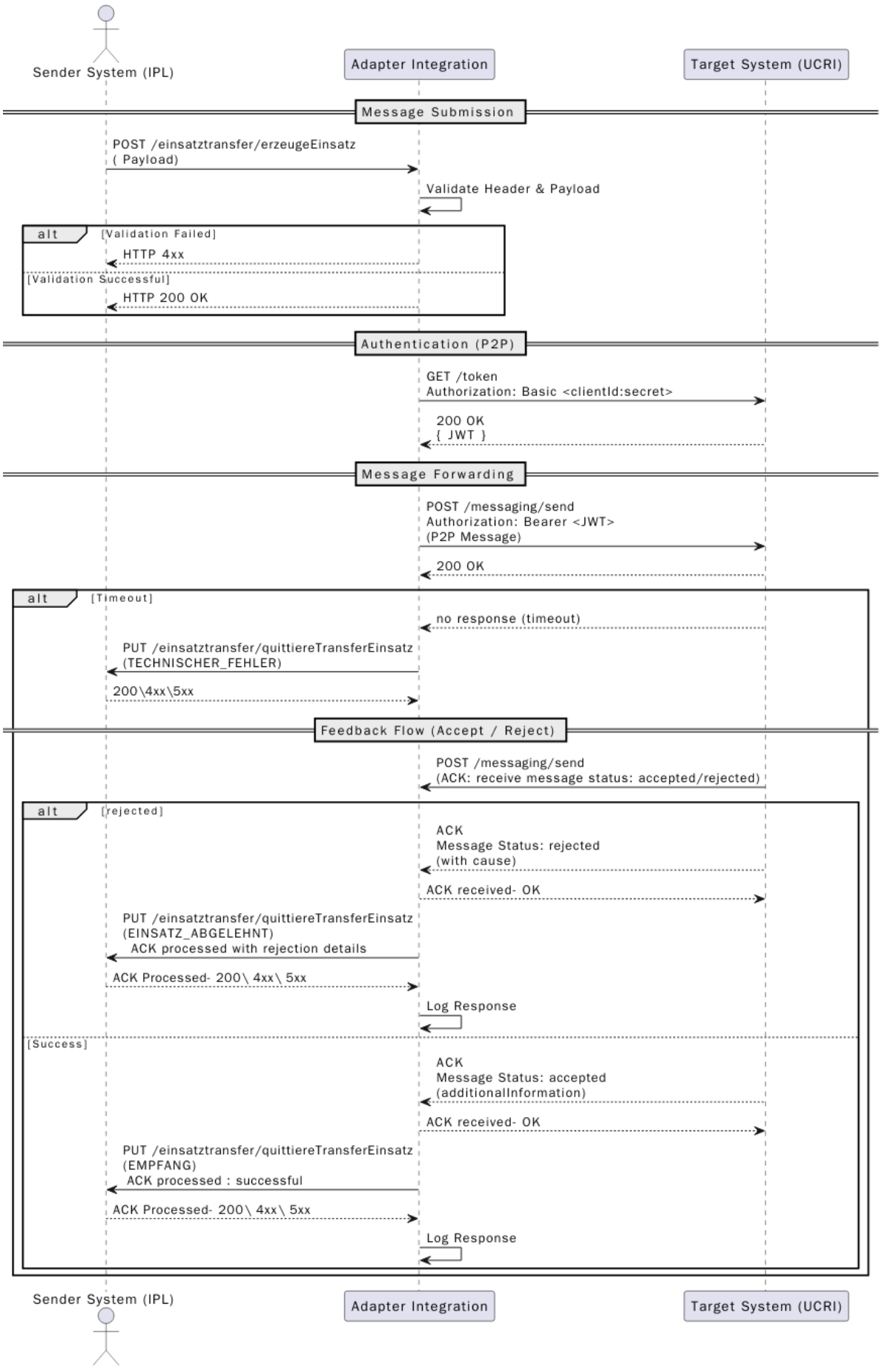
- The sender system submits a business payload to the Adapter Integration.
- The Adapter validates the incoming request and immediately responds with either a validation error or a successful acceptance.
- For valid requests, the Adapter authenticates against the target UCRM using the P2P `/token` endpoint.
- The Adapter forwards the message to the target UCRM via the `/messaging/send` P2P endpoint.
- The target UCRM processes the message and responds with acceptance or rejection based on UCRI-2.0 rules.
- The Adapter propagates the outcome back to the sender system, ensuring that message delivery and processing status are transparently communicated.

Open Question-

In the P2P flow, the same endpoint (POST /messaging/send) is used for sending and receiving messages.

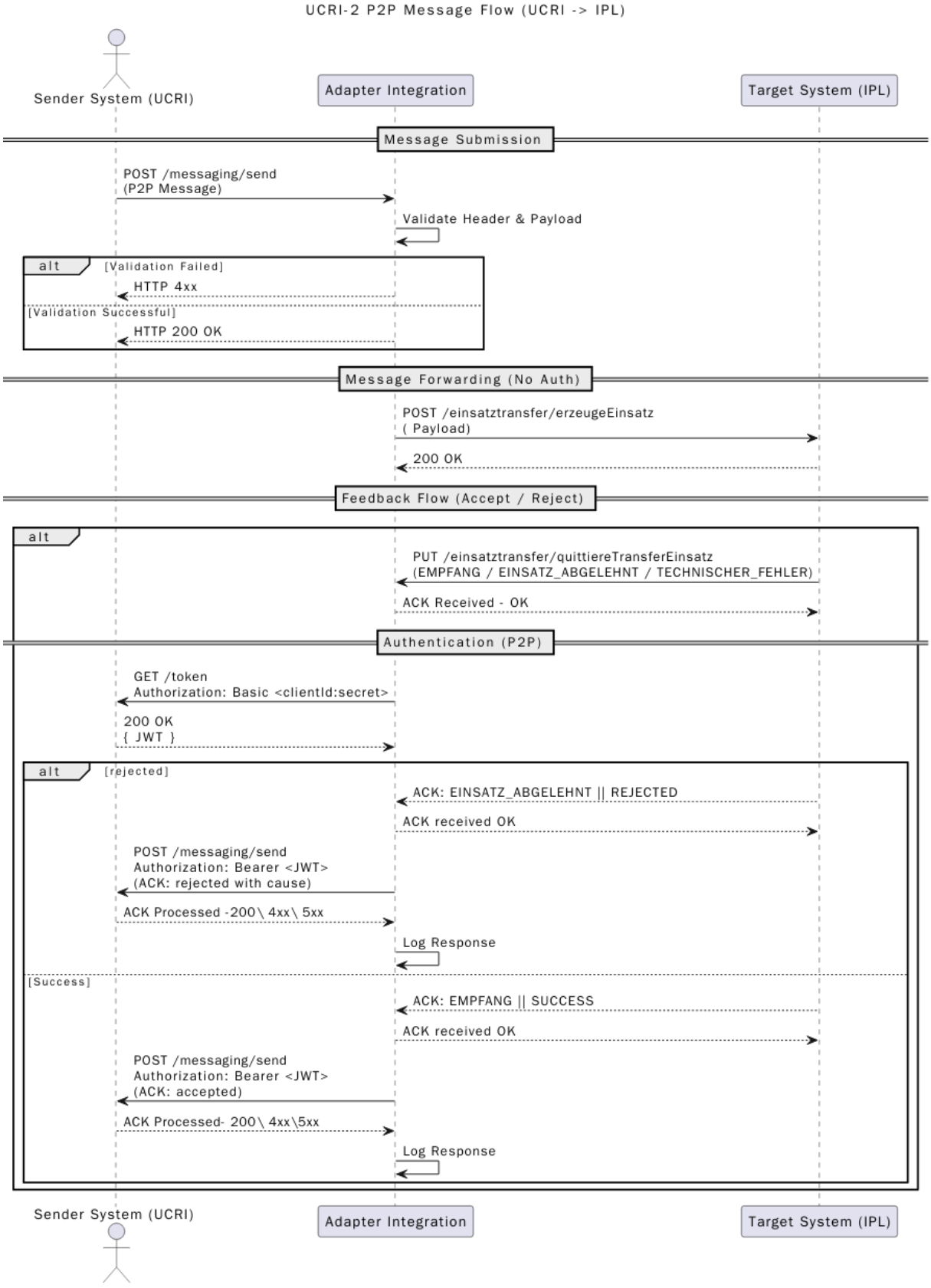


UCRI-2 P2P Message Flow (IPL -> UCRI)



Reverse Flow UCRI→IPL




UCRI-2 P2P Message Flow (UCRI -> IPL)
Sender System (UCRI)
Adapter Integration
Target System (IPL)
Message Submission
POST /messaging/send
(P2P Message)
Validate Header & Payload
alt
[Validation Failed]
HTTP 4xx
[Validation Successful]
HTTP 200 OK
Message Forwarding (No Auth)
POST /einsatztransfer/erzeugeEinsatz
(Payload)
200 OK
Feedback Flow (Accept / Reject)
alt
PUT /einsatztransfer/quittiereTransferEinsatz
(EMPFANG / EINSATZ_ABGELEHNT / TECHNISCHER_FEHLER)
ACK Received - OK
Authentication (P2P)
GET /token
Authorization: Basic <clientId:secret>
200 OK
{ JWT }
alt
[rejected]
ACK: EINSATZ_ABGELEHNT || REJECTED
ACK received OK
POST /messaging/send
Authorization: Bearer <JWT>
(ACK: rejected with cause)
ACK Processed -200\ 4xx\ 5xx
Log Response
[Success]
ACK: EMPFANG || SUCCESS
ACK received OK
POST /messaging/send
Authorization: Bearer <JWT>
(ACK: accepted)
ACK Processed- 200\ 4xx\5xx
Log Response
Sender System (UCRI)
Adapter Integration
Target System (IPL)